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Quaternion-Loop Quantum Gravity

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Abstract

It is shown that the Riemannian curvature of the 3-dimensional hypersurfaces in space-time, described by the Wilson loop integral, can be represented by a quaternion quantum operator induced by the SU(2) gauge potential, thus providing a justification for quaternion quantum gravity at the Tev energy scale.

Contents

1 Introduction

In one of his earliest attempts towards a canonical formulation of the gravitational field, Dirac described the propagation of 3-dimensional space-like hypersurfaces in space-time along an orthogonal time-like direction. The resulting non-zero Hamiltonian allowed the construction of the canonical equations, whose solutions describe the evolution of a special space-time foliation given by by 3-dimensional space-like hypersurfaces [1]. Since that procedure is not compatible with the diffeomorphism invariance of general relativity, the next logical step was to define a covariant foliation. For that purpose Arnowitt, Deser and Misner considered an arbitrary propagation direction in space-time, with a time-like lapse component N_0, and three space-line components N_i composing a shift vector [2]. Unfortunately, the presence of the shift vector implies that the Hamiltonian vanishes, thus frustrating the intended covariant canonical formulation. The problem persists even after applying Dirac's procedure for constrained systems [3], essentially because the Poisson bracket structure is not invariant under the group of diffeomorphisms [4].

In view of such negative results, it seemed for a while that a covariant canonical quantum gravity program describing the quantum fluctuations of 3-hypersurfaces in space-time was not possible. However an interesting alternative was proposed by A. Ashtekar, suggesting that a quantized $SU(2)$ gauge field acting as an auxiliary variable, may induce the quantization of the gravitational field [5]. Indeed, the Riemann curvature of the 3-dimensional hypersurfaces can be written in terms of the affine connection of the group of holonomy of triads along a closed loop, as given by the Wilson loop integral [6, 7]. Since the holonomy group is isomorphic to $SO(3)$, which in turn is isomorphic to the gauge group $SU(2)$, then the quantized $SU(2)$ gauge field induces the quantization

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of the 3-curvature. However, the mathematical difficulties associated with the expansion of the Wilson integral including the spin operators after the quantization (ie the Mandelstam inequalities), eventually led to a different route: Since the $SU(2)$ operators act on two-component spinor fields, then the curvature of the 3-dimensional hypersurfaces can also be expressed in terms of an underlying projective space defined at the Planck scale, as a discrete structure such as Penrose's spin network. The present stage of the development of loop quantum gravity, based on such spin network and using finite difference equations, is a current mainstream on quantum gravity [7, 8].

The purpose of this paper is to present an alternative to loop quantum gravity, based on the fact that the expression of the 3-dimensional Riemann curvature in terms of the Wilson integral is independent of the spinor discrete structure. We use the fact that the quaternion algebra is more fundamental than any of its representations, including the spinor representations. This last property allows us to make an explicit association between the curvature given by the Wilson loop integral and quaternion quantum mechanics, naturally suggesting the name of quaternion-loop quantum gravity. The main difference with the standard loop quantum gravity being that instead of using a discrete projective space structure, quaternion-loop quantum gravity is based on the standard continuum analysis of manifolds. In the next section we give a very brief review of properties of quaternion and its 2-component spinor representation. In section 3 we describe the curvature of a 3-dimensional hypersurface in terms of the Wilson loop integral and use the quantization of the $SU(2)$ gauge field to express that curvature as a quaternion quantum operator. At the end we give the example of the 3-dimensional curvatures induced by instanton and anti-instanton gauge fields.

2 Quaternions and Spinors

A common textbook explanation on the necessity of the complex formulation of quantum mechanics is that it is required to guarantee the completeness of the spectrum of eigenvalues of the Casimir spin operator [9]. However, since 2-component spinors are vectors belonging to the representation space of a representation of the group of automorphisms of the quaternion algebra, the above mentioned justification for complex quantum mechanics is really a justification for quaternion quantum mechanics. It has been suggested that the non-commutativity of the quaternion algebra may detail some properties of quantum fields and quantum geometry, which are not manifestly present in the simpler complex quantum theory [10, 11, 12].

The quaternion algebra is a Clifford algebra with two generators $\{e_1, e_2\}$ satisfying a multiplication table $e_ie_j + e_je_i = 2g_{ij}e_0, \quad e_0e_i = e_ie_0, \quad e_0e_0 = e_0$, where g_{ij} are the coefficients of a given quadratic form. It follows that the number of independent products of the generators is four, defining the dimension of the algebra. Denoting $e_3 = e_1e_2$, a general quaternion can be written as

$$X = X^0e_0 + X^1e_1 + X^2e_2 + X^3e_3$$

The multiplication table is invariant under the group of automorphisms $e'_\alpha = ue_\alpha u^{-1}$, which is isomorphic to the $SO(3)$ group. The conjugate of a quaternion is defined by $\bar{e}_i = -e_i, \quad \bar{e}_0 = e_0$, so that the real part of a quaternion is $Re(X) = (X + \bar{X})/2$, and the imaginary (or vector) part is $Im(X) = (X - \bar{X})/2$. Therefore, quaternions have a norm $||X||^2 = X\bar{X}$, and for $g_{ij} = \delta_{ij}$ the norm has the Euclidean expression

$$||X||^2 = X_0^2 + X_1^2 + X_2^2 + X_3^2$$

The existence of such norm allow us to construct the inverse of a quaternion as $X^{-1} = \bar{X}/||X||^2$. It also allows to write $||XY|| = ||X||||Y||$, a property of the quaternion algebra which is shared with the real and complex algebras. In fact these are the only known associative normed division algebras, a necessary condition for the definition of the standard mathematical analysis, including the notions of limit, continuity and derivative of a quaternion function $F(X)$ of a quaternion variable

X. While in general we have that the product of variations $\Delta F \Delta X^{-1} \neq \Delta X^{-1} \Delta F$, the division property says that

$$\lim_{\Delta X \to 0} ||\Delta F(X)(\Delta X)^{-1}|| = \lim_{\Delta X \to 0} ||(\Delta X)^{-1} \Delta F(X)|| = \lim_{\Delta X \to 0} \frac{||\Delta F(X)||}{||\Delta X||} \tag{1}$$

so that although the limit operations are well defined, the left and right derivatives need not be equal. If we impose that they are, then the analyticity conditions become too restrictive to allow for the definition of some relevant analytic functions [13].

The spinor representations of the quaternion algebra starts with the Pauli matrices

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0. \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{2}$$

which satisfy (for $g_{ij} = \delta_{ij}$) the same quaternion multiplication table $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}\sigma_0, \quad \sigma_0\sigma_i = \sigma_i\sigma_0, \quad \sigma_0\sigma_0 = \sigma_0$. In this representation a general quaternion corresponds to a 2x2 matrix

$$X = X^0\sigma_0 + X^1\sigma_1 + X^2\sigma_2 + X^3\sigma_3 = \begin{pmatrix} X^0 + X^3 & X^1 - iX^2 \\ X^1 + iX^2 & X^0 - X^3 \end{pmatrix} \tag{3}$$

Reciprocally, any 2x2 complex matrix operator acting on the 2-component spinor space can be also written as a quaternion operator. A quaternion X can also be expressed in terms of two-component spinors as $X = X^\mu \sigma_\mu = X^\mu \sigma_{\mu A}{}^{\bar{B}}\, \Psi^A \Psi_{\bar{B}}$ where the spin-tensor $\sigma_{\mu A}{}^{\bar{B}}$ are the components of each of the matrices σ_μ, Ψ_A are the components of the 2-component spinors and $\Psi^{\bar{A}}$ denotes its Hermitian conjugate. Therefore, a quaternion can be seen as determined by a more elementary spinor structure. Using a combinatorial analysis associated with the eigenvalues of the spin Casimir operator, Penrose postulated a discrete projective space, the spin network, which is a mathematical formulation of such underlying structure [14].

3 Quaternion-Loop Quantum Gravity

The curvature of the 3-dimensional hypersurface in the Riemann sense can be evaluated by transporting a tangent vector field along a closed loop consisting of a single continuous curve γ, starting at a point p and ending in the same point. Comparing the result with the original vector field at p, their difference can be expressed by the Wilson loop integral [12, 15]

$$R(\Gamma) = Pe^{\oint_\gamma \Gamma_i dx^i}, \quad i = 1..3 \tag{4}$$

where the coefficients Γ_i are the components of the affine connection ∇ evaluated in a reference frame (a triad) defined in the hypersurface. The motion of the triad along the loop characterizes the triad holonomy group which is isomorphic to $SO(3)$. The ordering factor P can be thought of as an integration constant. Admitting that Γ_i are continuous functions of the coordinates along the loop, the exponential in (4) is a well defined real analytic function represented by the standard exponential converging positive power series

$$e^{\oint_\gamma \Gamma_i dx^i} = 1 + \oint_\gamma \Gamma_i dx^i + (\oint_\gamma \Gamma_i dx^i)^2 + \cdots \tag{5}$$

The fact that the group of automorphisms of the quaternion algebra is also isomorphic to $SO(3)$, which is in turn isomorphic to $SU(2)$, suggests a classical-quantum correspondence between the curvature of the 3-dimensional hypersurfaces in space-time, expressed as a quaternion operator and the $SU(2)$ gauge field. Since Γ_i are the components of the connection of the triad holonomy group, we may establish this correspondence as

$$\Gamma_i \longleftrightarrow i\hbar \hat{\mathcal{A}}_i, \quad i = 1..3 \tag{6}$$

where $\hat{\mathcal{A}}_i$ are the 3-dimensional components of the quantized $SU(2)$ gauge potential. Replacing this in (4), we obtain the quantum curvature operator of the 3-dimensional hypersurface

$$\hat{\mathcal{R}}(\mathcal{A}) = \mathcal{P}e^{i\hbar\oint_\gamma \hat{\mathcal{A}}_i dx^i}, \quad i = 1..3 \tag{7}$$

However, as it was already commented, the analyticity of a quaternionic exponential function as in (7) is not well defined. Still, it is possible to apply (6) to each term of the classical expansion in the right hand side of (5), thus obtaining a series of quaternion polynomials, which converge to the quaternion exponential [13]:

$$e^{i\hbar\oint_\gamma \hat{\mathcal{A}}_i dx^i} \stackrel{def}{=} \sigma_0 + i\hbar\oint_\gamma \hat{\mathcal{A}}_i dx^i - \hbar^2(\oint_\gamma \hat{\mathcal{A}}_i dx^i)(\oint_\gamma \hat{\mathcal{A}}_j dx^j) + \cdots \tag{8}$$

In the sequence, we notice that the $SU(2)$ gauge potential is a connection 1-form written in the adjoint representation of the $SU(2)$ Lie algebra[1] as $\mathcal{A} = \mathcal{A}_\mu dx^\mu$. On the other hand this 1-form can also be written as a quaternion function in the Pauli basis as $\mathcal{A} = da_\mu\sigma^\mu$, so that $\mathcal{A}_\mu dx^\mu = da_\mu\sigma^\mu$, where da_μ is a 1-form and $\mathcal{A}_\mu$ are quaternion functions. The 3-dimensional curvature operator (7) can be written explicitly, using the 3-dimensional components $\mathcal{A}_i$ of the imaginary part of a quaternion function

$$Im\,\mathcal{A}(X) = \frac{1}{2}(f(X)dX - \overline{f(X)dX}) \tag{9}$$

Therefore, it is possible to to obtain the quantum curvature (7) by specifying a quaternion function expressed in the Pauli basis (2). As an example, consider the quaternion function

$$f(X) = \frac{\bar{X}}{1+|X|^2} \tag{10}$$

and its quaternion conjugate $\bar{f}(X) = X/(1+|X|^2)$, we may obtain self-dual (instanton) and anti-self-dual (anti-instanton) $SU(2)$ gauge fields respectively, defined by $\mathcal{F}^* = \pm\mathcal{F}$. In fact, replacing the above functions in (9) we obtain

$$Im\mathcal{A}(X) = \frac{1}{2}\left(\frac{\bar{X}dX - d\bar{X}X}{1+|X|^2}\right) \tag{11}$$

and the gauge curvature

$$\mathcal{F}(X) = \frac{d\bar{X}\wedge dX}{(1+|X|^2)^2} \tag{12}$$

This can be readily verified to satisfy the anti-self-dual (anti-instanton) condition [16]. More specifically, writing $X = x^\mu\sigma_\mu$ and $dX = dx^\mu\sigma_\mu$ in (11), then we obtain

$$\mathcal{A}(X) = \frac{-x^i dx^0 + x^0 dx^i}{1+|X|^2}\sigma_i = \mathcal{A}_0 dx^0 + \mathcal{A}_i dx^i \tag{13}$$

where

$$\mathcal{A}_0 = -\frac{x^i\sigma_i}{1+|X|^2} \text{ and } \mathcal{A}_i = \frac{x^0\sigma_i}{1+|X|^2} \tag{14}$$

are the components of the $SU(2)$ connection 1-form. Replacing the components $\mathcal{A}_i$ in (7) we obtain the quantum quaternion exponential curvature operator generated by an anti-instanton

$$\hat{\mathcal{R}}(\mathcal{A}) = \mathcal{P}e^{i\hbar\sigma_i\oint_\gamma \frac{x^0}{1+|X|^2}dx^i}, \quad i = 1..3 \tag{15}$$

By taking the quaternion conjugate of (10) we obtain the self-dual (instanton) solution.

[1] Using Greek indices for space-time, the corresponding Yang-Mills curvature is a 2-form $\mathcal{F} = \frac{1}{2}\mathcal{F}_{\mu\nu}\,dx^\mu\wedge dx^\nu$ where $\mathcal{F}_{\mu\nu} = [\mathcal{D}_\mu, \mathcal{D}_\nu]$, $D_\mu = \partial_\mu + \mathcal{A}_\mu$. The potential $\mathcal{A}_\mu$ is a solution of the Yang-Mills equations $\mathcal{D}\wedge\mathcal{F}^* = -4\pi J^*$ and $\mathcal{D}\wedge\mathcal{F} = 0$, where $\mathcal{F}^*_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma}\mathcal{F}^{\rho\sigma}$ denotes the components of the dual $\mathcal{F}^*$.

Concluding Remarks

In general relativity the observables of the gravitational field are associated with the eigenvalues of the curvature tensor. Therefore, the quaternion expression (7) may represent an observable effect of the gravity induced by the quantum $SU(2)$ gauge field. Such quantum effects should be observed at the same electro-weak energy level of the generating gauge potential (at the Tev scale), so that in principle it can be experimentally verified at the LHC in the form of small variations of the Minkowski zero curvature at the quantum scale. However, such result strongly contrasts with the hypothesis that the discrete structure of the standard loop quantum gravity is defined at the Planck scale.

As it was noted, quaternion loop quantum gravity differs fundamentally from the standard loop quantum gravity, because in the quaternion approach the existence of a classical background, the space-time geometry, is given by Einstein's theory. In a sense the quaternion loop quantum gravity is a less ambitious program than the standard loop quantum gravity proposition. Here the classical dynamics of the gravitational field is given by Einstein equations, providing the foliation of space-time by 3-dimensional hypersurfaces. However, we also obtain a quantum gravitational component, given by the Wilson curvature (7) which is effective only at the quantum scale, as derived from the time dependent $SU(2)$ Yang-Mills gauge field equations.

As a final remark, we may add that the quaternion loop quantum gravity does not depend on the construction of a gravitational Hamiltonian or, indeed of a traditional canonical formulation. Therefore it seems that it is possible to maintain the diffeomorphism invariance of the classical theory, even with the 3+1 decomposition of the space-time. This has been also suggested in the traditional loop quantum gravity, although admittedly it is far from obvious [17].

References

[1] P. A. M. Dirac, Proc.Roy.Soc.Lond. A246, 333, (1958).

[2] R. Arnowitt, S. Deser and C. Misner, *The Dynamics of General Relativity.* In *Gravitation: An introduction to Current Research*, L. Witten, Ed. John Willey & Sons, p. 227, (1962)

[3] P. A. M. Dirac, Phys. Rev. 114, 924, (1959).

[4] K. V. Kuchar, *Canonical Quantum Gravity,* 13^{th} Intl. Conf. on General Relativity & Gravitation, Cordoba, Argentina (1992). Ibid. *Time and Interpretations of Quantum Gravity* 4^{th} Can. Conf. on general Relativity and relati. Astrophys. World. Scientific (1991).

[5] A. Ashtekar, Phys. Rev. Lett. 57, 2244 (19860

[6] A. Ashtekar & Jerzy Lewandowski, Class.Quant.Grav. 21, 53 (2004), gr-qc/0404018; A. Ashtekar, *Loop Quantum Gravity: Four Recent Advances and a Dozen Frequently Asked Questions* arXiv:0705.2222 [gr-qc]

[7] Lee Smolin, *An Invitation to loop quantum gravity*, in Quantum theory and symmetries, Cincinatti, 655-682 (2003), hep-th/0408048.

[8] M. Han & Y. Ma, Intl. Jour. Mod. Phys. D16, 1397, (2007)

[9] C. Chevalley, *The construction and study of Certain Important Algebras.* The Math. Soc. Japan (1955).

[10] D. Finkelstein, *Notes on Quaternion Quantum Mechanics*, Preprint CERN-Th 59-9 (1959), D. Finkelstein et all, Jour. Math. Phys. 3, 207 (1962), ibid 4, 788, (1963).

[11] S. L. Adler, *Quaternion Quantum Mechanics and Quantum Fields*, Oxford U.P. (1995).

[12] S. L. Adler, *Non-Adiabatic Geometric Phase in Quaternionic Hilbert Space*, Preprint IASSNS-HEP-96 /89 (1996).

[13] M. D. Maia & V.B. Bezerra, Int.J.Theor.Phys. 40, 1283, (2001), hep-th/0107107

[14] R. Penrose, *Angular Momentum: An approach to Combinatorial Space-time*, in *Quantum Theory and Beyond*, Ted Bastin, C.U.P. (1971)

[15] G. Modanese, Phys. Rev. D49, 6534 (1994)

[16] M. F. Atiyah, *Geometry of Yang-Mills Fields* Scuola Normale Superiore, Pisa (1979).

[17] A. Ashtekar, *Some surprising implications of background independence in Canonical Quantum Gravity*, arXiv:0904.0184v1.